

Git and GitHub

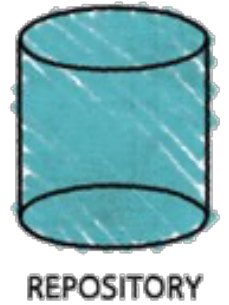
Programmazione Avanzata 2025-26

Version Control Systems

- Need to **record changes** to a file or a set of files over time so that one can **recall specific versions later**
- Three generations:
 - Local (RCS, SCCS)
 - Centralized (CVS, Subversion, Team Foundation Server)
 - Distributed (Mercurial, **Git**)

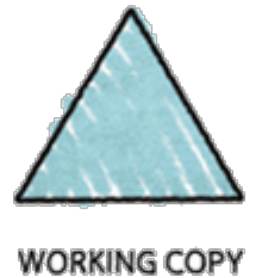
Basic concepts: Repository

- Place where the developer stores all his or her work
- Contains **every version** of such work that has ever existed
 - Files
 - Directories layout
 - History
- Can be **shared** with the whole team



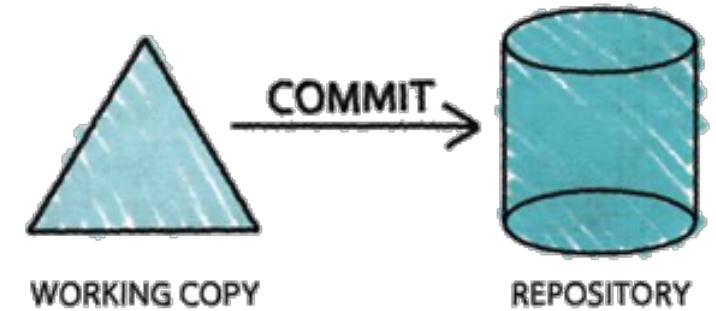
Basic concepts: Working copy

- A **snapshot** (part) of the repository used for... working
- Place **where changes happens**
- Private, **not shared** with the team
- It also contains some metadata so that it can keep track of the state of things
 - Has a file been modified?
 - Is this file new?
 - Has a file been deleted?



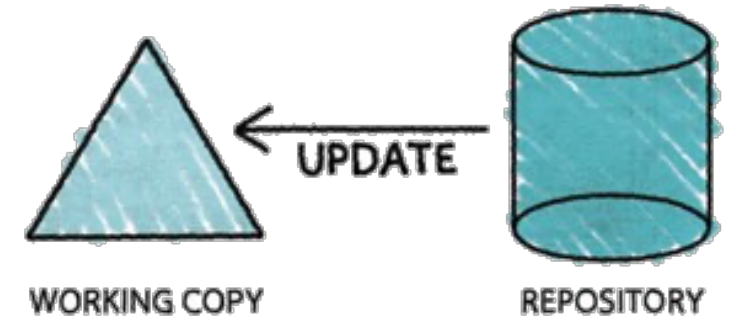
Basic concepts: Commit

- The operation that actually modifies the repository
 - Performed “atomically” by modern Version Control Systems
 - The integrity of the repository is ensured
- The developer typically provides a log message (or comment) when making commit
 - To explain the changes he or she has made
 - The message becomes part of the history of the repository



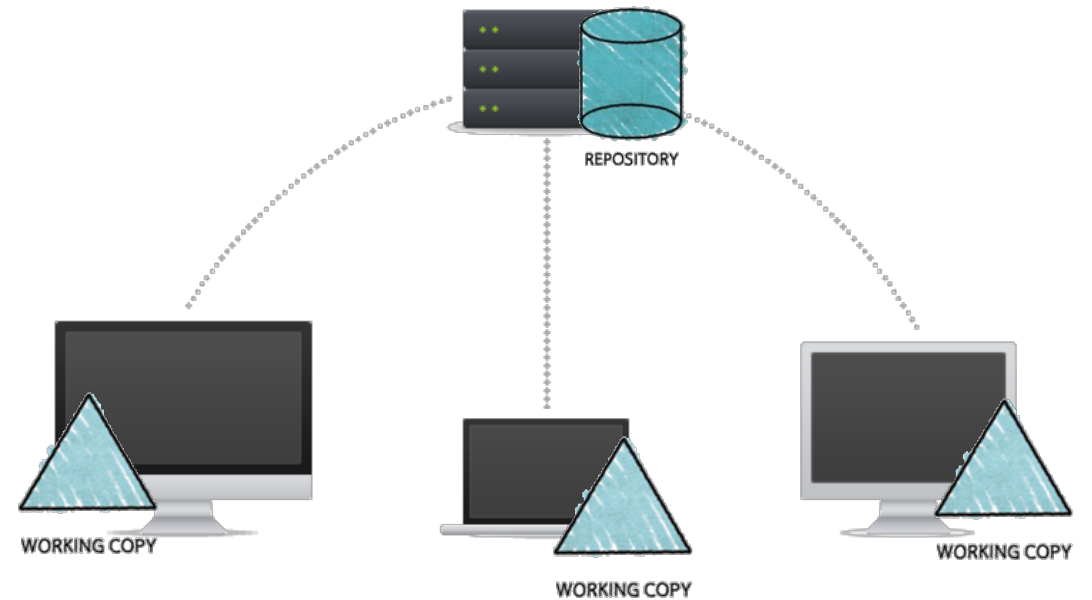
Basic concepts: Update

- The operation that updates the working copy with respect to (based on) the repository
 - Applies changes from the repository
 - Merges such changes with the ones the developer has made to his or her working copy, if necessary



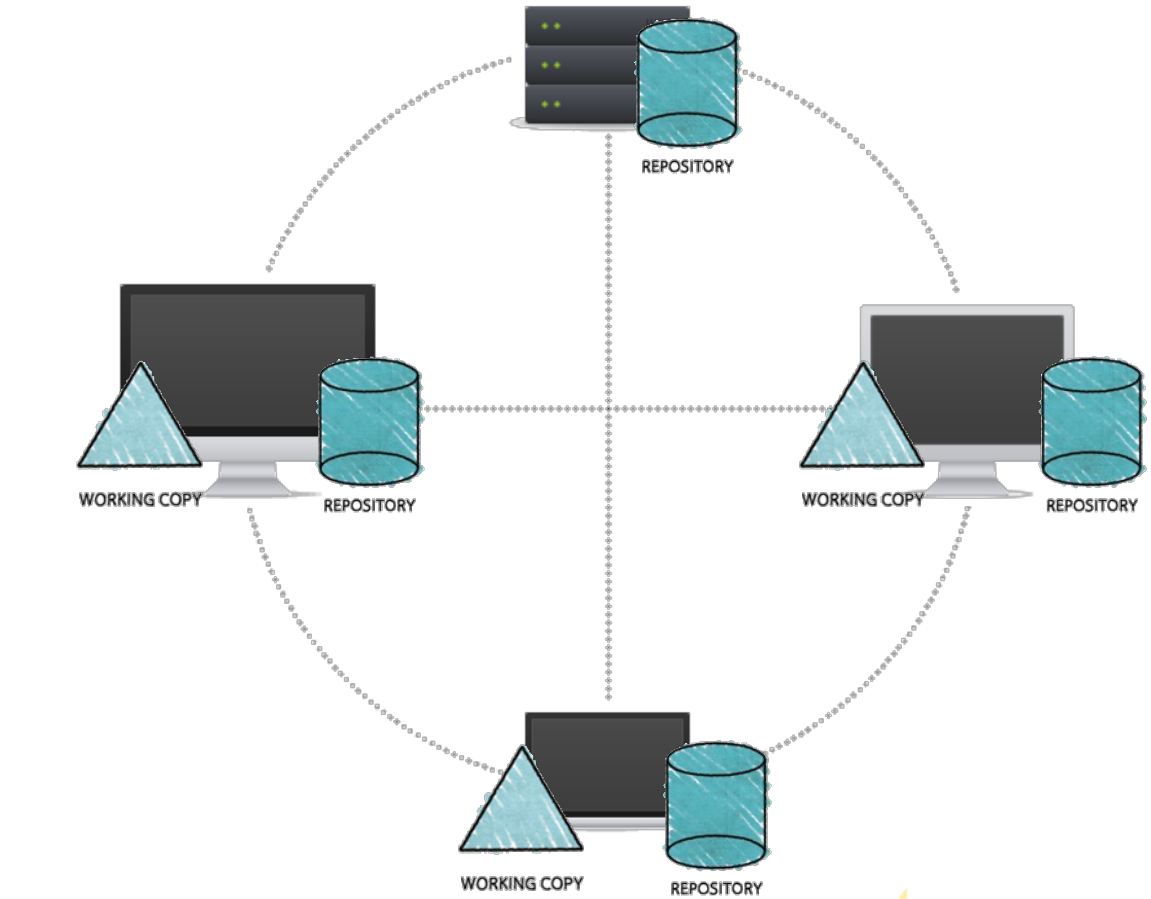
Centralized Version Control

- One central repository
- Client-server relationship



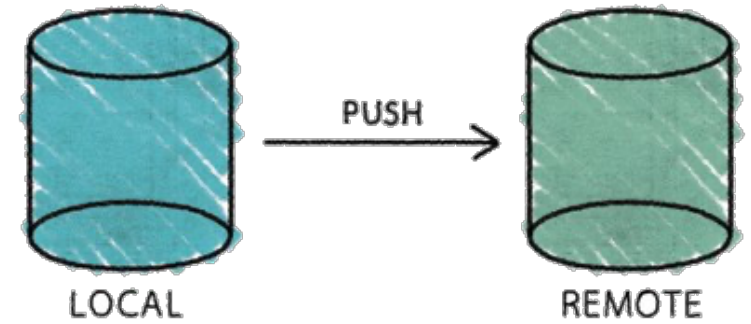
Distributed Version Control

- Clients and server have the full copy of the repository
 - Local repositories “clone” a remote repository
 - It is possible to have more than one server
 - A mechanism to synchronize repositories is needed



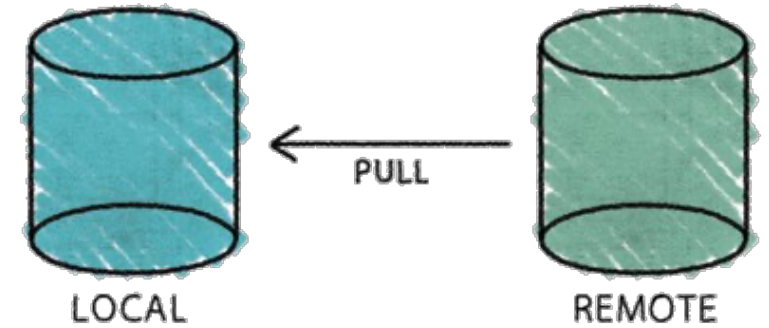
More basic cocepts: Push

- Operation that copies changesets from a local repository instance to a remote one
 - Implements the synchronization between two repository instances



More basic concepts: Pull

- Operation that copies changesets from a remote repository instance to a local one
 - Implements the other direction of the synchronization between two repository instances



Git

- Distributed Version Control System
- Born
 - On 2005 for the Linux kernel project
 - To be used via command line
- Website: <http://git-scm.com>
- Highlights:
 - Free and open source
 - Strong support for non-linear development
 - Fully distributed
 - Efficient handling of large projects
 - Cryptographic authentication of history



Getting started with Git

- Standard installations
 - <http://git-scm.com/downloads>
- Available for all the platforms
- Git Graphical Applications
 - <http://git-scm.com/downloads/guis>
- Suggestions
 - GitExtensions, SourceTree, Fork
- Git is already integrated in PyCharm

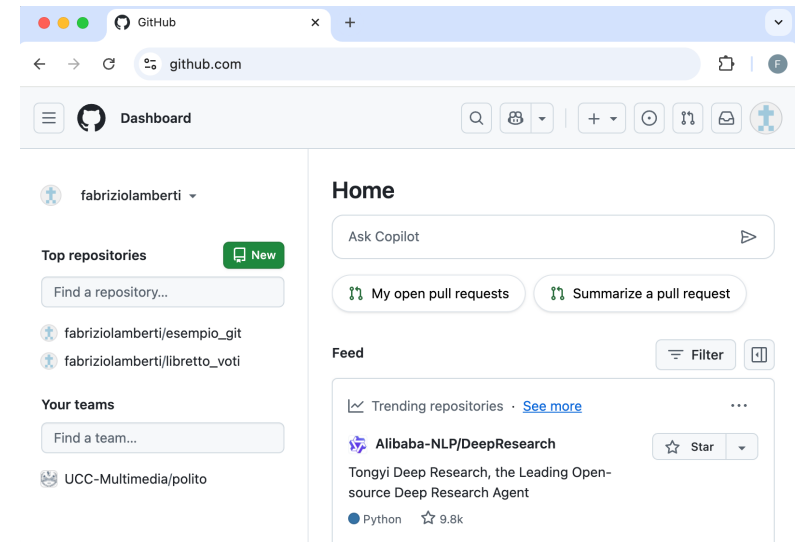


Hosted Git

- To have (at least) one remote repository
 - Set up owned Git server, complicated
- Popular cloud services
 - GitHub, <https://github.com/>
 - Bitbucket, <https://bitbucket.org/>
 - GitLab, <https://about.gitlab.com/gitlab-com/>

GitHub

- Slightly different than other code-hosting sites
 - Instead of being primarily based on the project, it is user-centric
 - Social coding
- Owned by Microsoft
 - Free account to host as many open source project as one wants
- Free plans for students
 - <https://education.github.com>



Using Git and GitHub with PyCharm

- Create a project in PyCharm and push it on GitHub
 1. In PyCharm, “File/New Project...”
 - Tick “Create Git Repository”
 2. Create and edit Python files
 - Add files to Git repository, if asked
 3. Create a new (empty) project in GitHub
 - Copy the Project URL from GitHub
 4. In PyCharm “Commit and Push...” to push changes
 - The first time it is necessary to use “Define Remote”, inserting the URL

Using Git and GitHub with PyCharm

- Work on a project that is on GitHub (from other developer)
 1. Fork it from GitHub webpage, creating a copy in own GitHub account (with writing privileges)
 2. In PyCharm, “Clone” it in local repository
 3. Work on the project
 4. Commit and push the changes

Branches

- Used to develop features isolated from each other
 - The main (or master) branch is the “default” branch when a repository is created
 - Other branches can be used for development and merged back to the master branch upon completion
- Branches can be local (in local repository) or pushed to a remote repository

References

- Git Reference
 - <http://gitref.org/>
- Git – The simple guide
 - <http://rogerdudler.github.io/git-guide/>
- Git Documentation
 - <http://git-scm.com/docs>
- Pro Git (online book)
 - <http://git-scm.com/book>
- Version Control by Example (online book)
 - <http://www.ericSink.com/vcbe/>

References

- Try Git!
 - <http://try.github.io/>
- Various Git resources
 - <https://help.github.com/articles/what-are-other-good-resources-for-learning-git-and-github>
- A successful Git branching model
 - <http://nvie.com/posts/a-successful-git-branching-model/>
- Some Git (graphical) clients
 - <http://git-scm.com/downloads/guis>